

Mode 8

$$\text{DP16 } (16 \times 16) \rightarrow \text{DP8 } (16 \times 16) \rightarrow \text{DP8 } (16 \times 8) \rightarrow \text{DP8 } (16 \times 4) \rightarrow \text{DP8 } (8 \times 4)$$

Step 1 — DP16 (16×16)

$$\text{DP16 } (16 \times 16) = \sum_{i=0}^{15} a_i b_i$$

Step 2 — DP8 (16×16)

$$\text{DP16 } (16 \times 16) = \underbrace{\sum_{i=0}^7 a_i b_i}_{\text{DP8 } (16 \times 16)} + \underbrace{\sum_{i=8}^{15} a_i b_i}_{\text{DP8 } (16 \times 16)}$$

Step 3 — DP8 (16×8)

$$\text{DP16 } (16 \times 16) = 2^8 \underbrace{\sum_{i=0}^7 a_i b_i^{hi}}_{\text{DP8 } (16 \times 8)} + \underbrace{\sum_{i=0}^7 a_i b_i^{lo}}_{\text{DP8 } (16 \times 8)} + 2^8 \underbrace{\sum_{i=8}^{15} a_i b_i^{hi}}_{\text{DP8 } (16 \times 8)} + \underbrace{\sum_{i=8}^{15} a_i b_i^{lo}}_{\text{DP8 } (16 \times 8)}$$

Step 4 — DP8 (16×4)

$$\begin{aligned} \text{DP16 } (16 \times 16) = & 2^{12} \underbrace{\sum_{i=0}^7 a_i b_i^{hh}}_{\text{DP8 } (16 \times 4)} + 2^8 \underbrace{\sum_{i=0}^7 a_i b_i^{hl}}_{\text{DP8 } (16 \times 4)} + 2^4 \underbrace{\sum_{i=0}^7 a_i b_i^{lh}}_{\text{DP8 } (16 \times 4)} + \underbrace{\sum_{i=0}^7 a_i b_i^{ll}}_{\text{DP8 } (16 \times 4)} \\ & + 2^{12} \underbrace{\sum_{i=8}^{15} a_i b_i^{hh}}_{\text{DP8 } (16 \times 4)} + 2^8 \underbrace{\sum_{i=8}^{15} a_i b_i^{hl}}_{\text{DP8 } (16 \times 4)} + 2^4 \underbrace{\sum_{i=8}^{15} a_i b_i^{lh}}_{\text{DP8 } (16 \times 4)} + \underbrace{\sum_{i=8}^{15} a_i b_i^{ll}}_{\text{DP8 } (16 \times 4)} \end{aligned}$$

Step 5 — DP8 (8×4)

$$\begin{aligned}
\text{DP16 } (16 \times 16) = & 2^{20} \underbrace{\sum_{i=0}^7 a_i^{hi} b_i^{hh}}_{\text{DP8 } (8 \times 4)} + 2^{20} \underbrace{\sum_{i=8}^{15} a_i^{hi} b_i^{hh}}_{\text{DP8 } (8 \times 4)} + 2^{16} \underbrace{\sum_{i=0}^7 a_i^{hi} b_i^{hl}}_{\text{DP8 } (8 \times 4)} + 2^{16} \underbrace{\sum_{i=8}^{15} a_i^{hi} b_i^{hl}}_{\text{DP8 } (8 \times 4)} \\
& + 2^{12} \underbrace{\sum_{i=0}^7 a_i^{lo} b_i^{hh}}_{\text{DP8 } (8 \times 4)} + 2^{12} \underbrace{\sum_{i=8}^{15} a_i^{lo} b_i^{hh}}_{\text{DP8 } (8 \times 4)} + 2^{12} \underbrace{\sum_{i=0}^7 a_i^{hi} b_i^{lh}}_{\text{DP8 } (8 \times 4)} + 2^{12} \underbrace{\sum_{i=8}^{15} a_i^{hi} b_i^{lh}}_{\text{DP8 } (8 \times 4)} \\
& + 2^8 \underbrace{\sum_{i=0}^7 a_i^{lo} b_i^{hl}}_{\text{DP8 } (8 \times 4)} + 2^8 \underbrace{\sum_{i=8}^{15} a_i^{lo} b_i^{hl}}_{\text{DP8 } (8 \times 4)} + 2^8 \underbrace{\sum_{i=0}^7 a_i^{hi} b_i^{ll}}_{\text{DP8 } (8 \times 4)} + 2^8 \underbrace{\sum_{i=8}^{15} a_i^{hi} b_i^{ll}}_{\text{DP8 } (8 \times 4)} \\
& + 2^4 \underbrace{\sum_{i=0}^7 a_i^{lo} b_i^{lh}}_{\text{DP8 } (8 \times 4)} + 2^4 \underbrace{\sum_{i=8}^{15} a_i^{lo} b_i^{lh}}_{\text{DP8 } (8 \times 4)} + \underbrace{\sum_{i=0}^7 a_i^{lo} b_i^{ll}}_{\text{DP8 } (8 \times 4)} + \underbrace{\sum_{i=8}^{15} a_i^{lo} b_i^{ll}}_{\text{DP8 } (8 \times 4)}
\end{aligned}$$